CSE 5710 - Data Mining for Data Science

Data Mining Introduction and Foundational Concepts

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   - Differences from Machine Learning
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Definition - Data Mining

Definition

Data mining is the process of exploring data in order to detect patterns and extract insights.

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Definition - Machine Learning

Definition

Machine Learning is a branch of artificial intelligence concerned with developing algorithms for learning from data for the primary purpose of making predictions on *new* data.

Comparison of Data Mining and Machine Learning

1. Focus
   - DM - Focus is on uncovering hidden patterns
   - ML - Focus is on making predictions
2. Predictive Power
   - DM - primarily concerned with using existing data to gain insights
   - ML - primarily concerned with making predictions on *new data*.
3. Learning Process
   - DM - typically involves predefined rules and algorithms to identify patterns
   - ML - algorithms can adapt and improve as new data is made available.

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Algebra

- When formalizing intuitive concepts, a common apporach is to construct a set of objects (symbols) and a set of rules to manipulate these objects.
- Such a system is called an *algebra*.

Linear Algebra

Definition

Linear algebra is the study of *vectors* and the rules to manipulate them.

Vector

- A *vector* is a concise representation of a complex object in n-dimensional space, $\mathbb{R}^n$.
- A *complex object* is one characterized by *multiple* attributes as opposed to a simple (scalar) object, defined by only a *single* value.
- Note that all n attributes in a *vector* are numeric (as opposed to categorical).
- In this course, we will represent all vectors as a bold letter, e.g., $\boldsymbol{x}$ as opposed to $\vec{x}$.

Example - Person object

Example - Person objects

person_1 = (age, height, salary) $\Rightarrow$ All features numeric $\Rightarrow$ vector
person_2 = (ethnicity, age, height, salary) $\Rightarrow$ *NOT* all features numeric $\Rightarrow$ NOT a vector

Attributes in the Problem Domain

Attributes in the problem domain may be classified into two main types

- **Numeric Attributes**: real-valued or integer-valued domain
  - *Interval-scaled*: only differences are meaningful
    e.g., `temperature`
  - *Ratio-scaled*: differences and ratios are meaningful
    e..g, `Age`
- **Categorical Attributes**: set-valued domain composed of a set of symbols
  - *Nominal*: only equality is meaningful
    e.g., domain(`Sex`) = { `M, F`}
  - *Ordinal*: both equality (are two values the same?) and inequality (is one value less than another?) are meaningful
    e.g., domain(`Education`) = { `High School`, `BS`, `MS`, `PhD`}

Vector features vs. Problem Domain Attributes

- Many (if not most) data mining algorithms require we convert any/all categegorical attributes to numeric attributes.
- E.g. Clustering algorithms based on *distance between objects (vectors)* since calculating distance requires all attributes to be numeric.

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Matrix

- *Matrix*: a concise representation for a collection of complex objects each of which is represented as a *vector* and for which we have well-defined rules for manipulation.
- An $n \times d$ *data matrix*, with n rows and d columns is given as

$$\boldsymbol{D} = \left(\begin{array}{c|cccc} & X_1 & X_2 & \cdots & X_d \\ \hline \boldsymbol{x}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \boldsymbol{x}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{array}\right)$$

Matrix

- An $n \times d$ *data matrix*, with n rows and d columns is given as

$$\boldsymbol{D} = \left(\begin{array}{c|cccc} & X_1 & X_2 & \cdots & X_d \\ \hline \boldsymbol{x}_1 & x_{11} & x_{12} & \cdots & x_{1d} \\ \boldsymbol{x}_2 & x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{x}_n & x_{n1} & x_{n2} & \cdots & x_{nd} \end{array}\right)$$

- **Rows**: *instances*, *examples*, *records*, *transactions*, *objects*, *points*, *feature-vectors*, etc. Given as a d-tuple

$$\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{id}) \in \mathbb{R}^d$$

- **Columns**: *attributes*, *properties*, *features*, *dimensions*, *variables*, *fields*, etc. Given as an n-tuple

$$X_j = (x_{1j}, x_{2j}, \ldots, x_{nj}) \in \mathbb{R}^n$$

Example: Iris Dataset Extract

Example: Extract from the Iris Dataset

	Sepal length X_1	**Sepal width** X_2	**Petal length** X_3	**Petal width** X_4	**Class** X_5
$\boldsymbol{x}_1$	5.9	3.0	4.2	1.5	Iris-versicolor
$\boldsymbol{x}_2$	6.9	3.1	4.9	1.5	Iris-versicolor
$\boldsymbol{x}_3$	6.6	2.9	4.6	1.3	Iris-versicolor
$\boldsymbol{x}_4$	4.6	3.2	1.4	0.2	Iris-setosa
$\boldsymbol{x}_5$	6.0	2.2	4.0	1.0	Iris-versicolor
$\boldsymbol{x}_6$	4.7	3.2	1.3	0.2	Iris-setosa
$\boldsymbol{x}_7$	6.5	3.0	5.8	2.2	Iris-virginica
$\boldsymbol{x}_8$	5.8	2.7	5.1	1.9	Iris-virginica
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\boldsymbol{x}_{149}$	7.7	3.8	6.7	2.2	Iris-virginica
$\boldsymbol{x}_{150}$	5.1	3.4	1.5	0.2	Iris-setosa

Example: Iris Dataset Extract - Projection in 2D and 3D

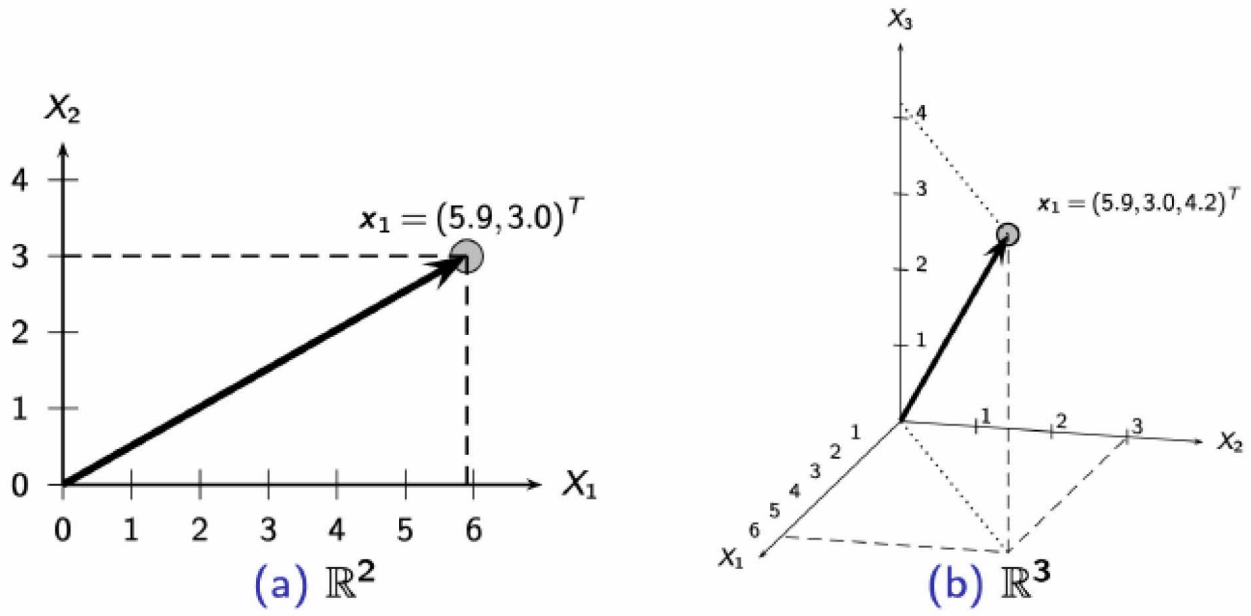
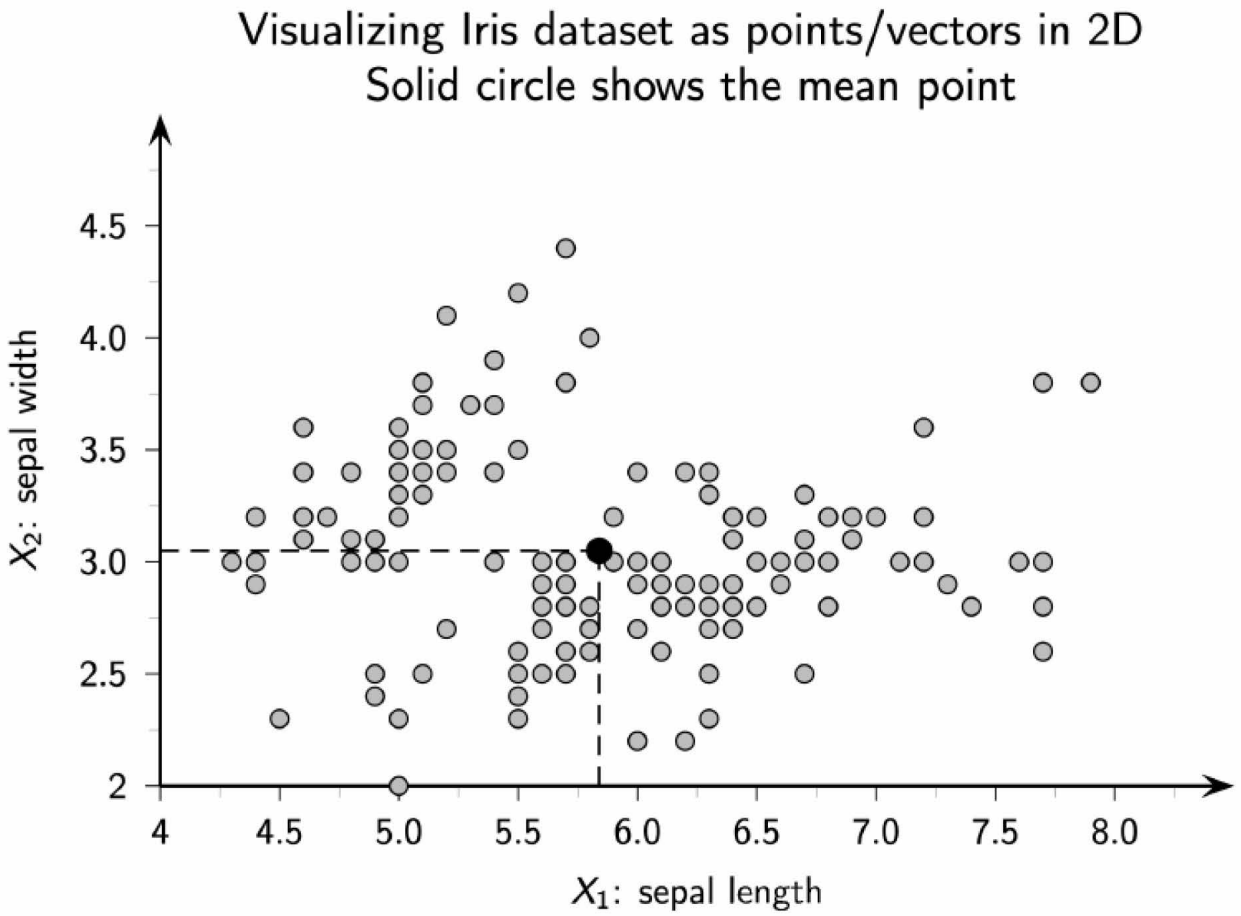


Figure: Projections of $\boldsymbol{x}_1 = (5.9, 3.0, 4.2, 1.5)^T$ in 2D and 3D

Example: Iris Dataset Extract - Projection in 2D



Matrix Representation Formats

$$\boldsymbol{D} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix} = \begin{pmatrix} - \ \boldsymbol{x}_1^T \ - \\ - \ \boldsymbol{x}_2^T \ - \\ \vdots \\ - \ \boldsymbol{x}_n^T \ - \end{pmatrix} = \begin{pmatrix} | & | & & | \\ X_1 & X_2 & \cdots & X_d \\ | & | & & | \end{pmatrix}$$

Mean of Matrix

The *mean* of the data matrix $\boldsymbol{D}$ is the average of all the points:

$$mean(\boldsymbol{D}) = \boldsymbol{\mu} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \in \mathbb{R}^d$$

$\boldsymbol{\mu}$ is $(d \times 1)$ vector whose values are the averages of the d attributes across the n objects.

Centered Matrix

The *centered matrix* is found by subtracting the mean from all the points:

$$\overline{\boldsymbol{D}} = \boldsymbol{D} - \mathbf{1} \cdot \boldsymbol{\mu}^T = \begin{pmatrix} \boldsymbol{x}_1^T \\ \boldsymbol{x}_2^T \\ \vdots \\ \boldsymbol{x}_n^T \end{pmatrix} - \begin{pmatrix} \boldsymbol{\mu}^T \\ \boldsymbol{\mu}^T \\ \vdots \\ \boldsymbol{\mu}^T \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_1^T - \boldsymbol{\mu}^T \\ \boldsymbol{x}_2^T - \boldsymbol{\mu}^T \\ \vdots \\ \boldsymbol{x}_n^T - \boldsymbol{\mu}^T \end{pmatrix} = \begin{pmatrix} \overline{\boldsymbol{x}}_1^T \\ \overline{\boldsymbol{x}}_2^T \\ \vdots \\ \overline{\boldsymbol{x}}_n^T \end{pmatrix}$$

where $\overline{\boldsymbol{x}}_i = \boldsymbol{x}_i - \boldsymbol{\mu}$ is a centered point, and $\mathbf{1} \in \mathbb{R}^n$ is the vector of ones.

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Dot Product of 2 Vectors

Given two points $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{R}^m$, the *dot product* of $\boldsymbol{a}$ and $\boldsymbol{b}$ is as follows:

$$\begin{aligned}\boldsymbol{a}^T\boldsymbol{b} &= a_1b_1 + a_2b_2 + \cdots + a_mb_m \\ &= \sum_{i=1}^{m} a_ib_i\end{aligned}$$

Euclidian Norm of Vector

The *Euclidean norm* or *length* of a vector $\boldsymbol{a}$ is defined as:

$$\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a}^T\boldsymbol{a}} = \sqrt{\sum_{i=1}^{m} a_i^2}$$

The *unit vector* in the direction of $\boldsymbol{a}$ is $\boldsymbol{u} = \frac{\boldsymbol{a}}{\|\boldsymbol{a}\|}$ with $\|\boldsymbol{u}\| = 1$.

Distance between Vectors

The *distance* between vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is given as

$$\|\boldsymbol{a} - \boldsymbol{b}\| = \sqrt{\sum_{i=1}^{m} (a_i - b_i)^2}$$

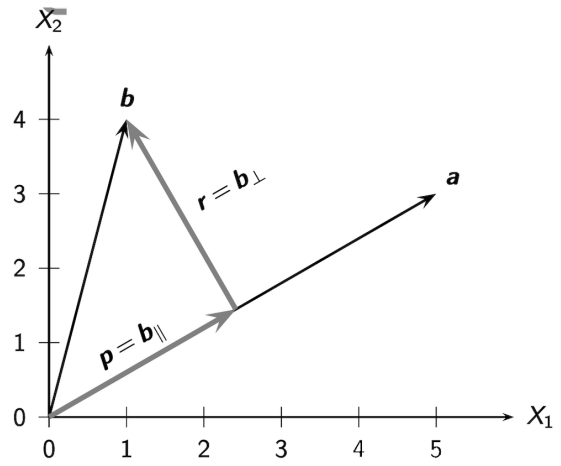
Orthogonality

Definition

Two vectors, $\boldsymbol{x}$ and $\boldsymbol{y}$ are **orthogonal** or **perpendicular** if $\boldsymbol{x} \cdot \boldsymbol{y} = 0$.
Notation: $\boldsymbol{x} \perp \boldsymbol{y}$ means the same as $\boldsymbol{x} \cdot \boldsymbol{y} = 0$.

Orthogonal Projection

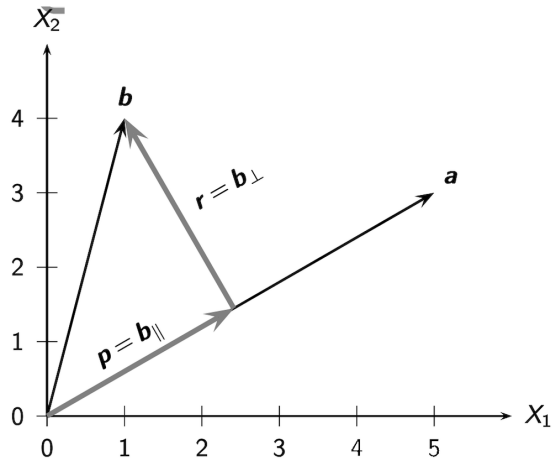
Suppose we have the following pair of vectors $\boldsymbol{a}$ and $\boldsymbol{b}$:



Note that $\boldsymbol{r} \perp \boldsymbol{a}$ and that $\boldsymbol{r} = \boldsymbol{b} - \boldsymbol{p}$.

Suppose further we wish to calculate the coordinates of vector, $\boldsymbol{p}$ which is a projection of vector $\boldsymbol{b}$ onto vector $\boldsymbol{a}$. (Note that $\boldsymbol{p}$ is also denoted $\boldsymbol{b}_\parallel$ since it is paralel to $\boldsymbol{a}$.)

Orthogonal Projection



- First, we note that $\boldsymbol{p} = c\boldsymbol{a}$ where c is a scalar constant.
- We also know that since $\boldsymbol{r} = (\boldsymbol{b} - \boldsymbol{p})$ and $\boldsymbol{r} \perp \boldsymbol{a}$, that $(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} = 0$ by orthogonality.

Orthogonal Projection

So, we have:

Orthogonal Projection

So, we have:

$$(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} = 0$$

Orthogonal Projection

So, we have:

$$(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} = 0$$
$$(\boldsymbol{b} - c\boldsymbol{a}) \cdot \boldsymbol{a} = 0$$

Orthogonal Projection

So, we have:

$$
\begin{aligned}
(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} &= 0 \\
(\boldsymbol{b} - c\boldsymbol{a}) \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} - c\boldsymbol{a} \cdot \boldsymbol{a} &= 0
\end{aligned}
$$

Orthogonal Projection

So, we have:

$$
\begin{aligned}
(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} &= 0 \\
(\boldsymbol{b} - c\boldsymbol{a}) \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} - c\boldsymbol{a} \cdot \boldsymbol{a} &= 0 \\
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\end{aligned}
$$

Orthogonal Projection

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\boldsymbol{b} \cdot \boldsymbol{a} &= c\boldsymbol{a} \cdot \boldsymbol{a} \\
\boldsymbol{a} \cdot \boldsymbol{b} &= c\boldsymbol{a} \cdot \boldsymbol{a}
\end{aligned}
$$

Orthogonal Projection

So, we have:

$$
\begin{aligned}
(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} &= 0 \\
(\boldsymbol{b} - c\boldsymbol{a}) \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} - c\boldsymbol{a} \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} &= c\boldsymbol{a} \cdot \boldsymbol{a} \\
\boldsymbol{a} \cdot \boldsymbol{b} &= c\boldsymbol{a} \cdot \boldsymbol{a} \\
\therefore \quad c &= \frac{\boldsymbol{a} \cdot \boldsymbol{b}}{||\boldsymbol{a}||^2} = \frac{\boldsymbol{a}^T\boldsymbol{b}}{||\boldsymbol{a}||^2} = \frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}}
\end{aligned}
$$

Orthogonal Projection

So, we have:

$$
\begin{aligned}
(\boldsymbol{b} - \boldsymbol{p}) \cdot \boldsymbol{a} &= 0 \\
(\boldsymbol{b} - c\boldsymbol{a}) \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} - c\boldsymbol{a} \cdot \boldsymbol{a} &= 0 \\
\boldsymbol{b} \cdot \boldsymbol{a} &= c\boldsymbol{a} \cdot \boldsymbol{a} \\
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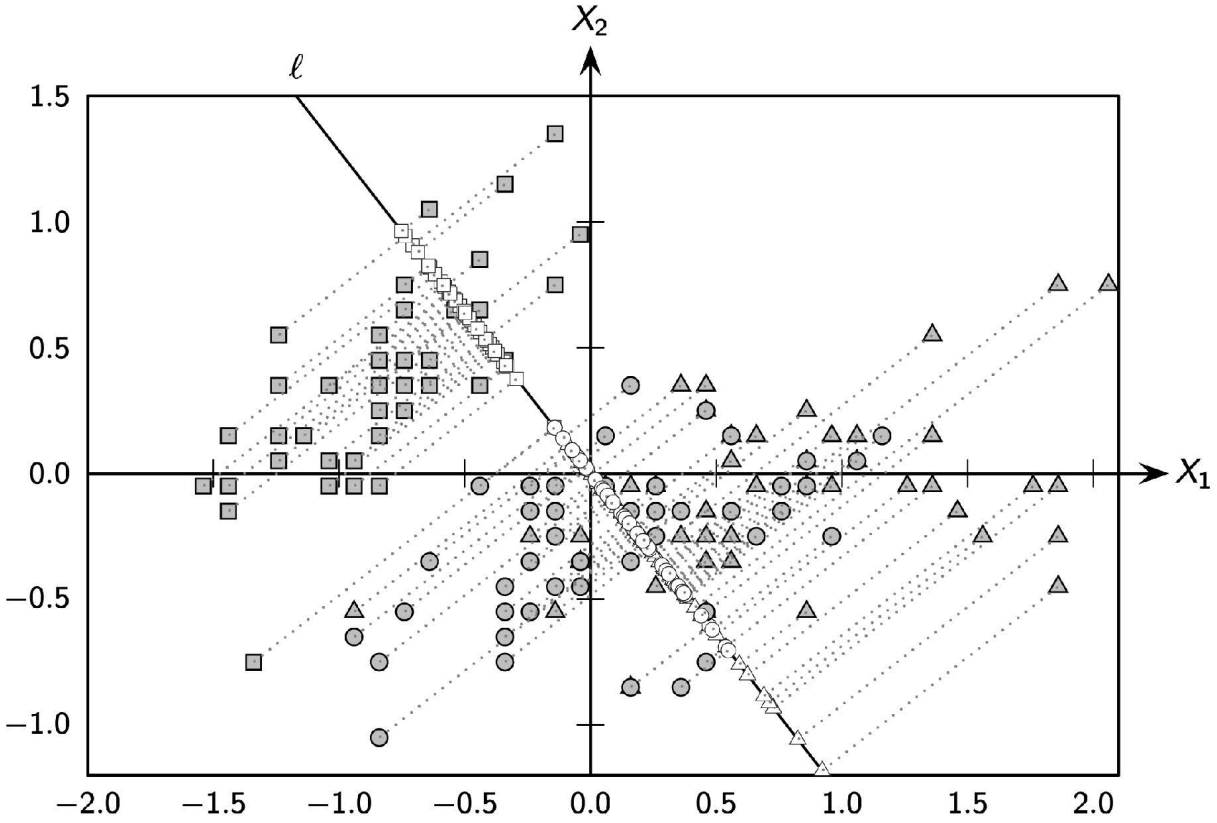
\end{aligned}
$$

So, we have:

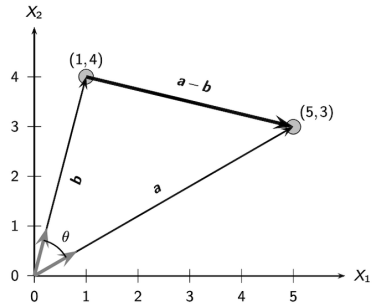
$$
\boxed{\boldsymbol{p} = c\boldsymbol{a} = \left(\frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}}\right)\boldsymbol{a}}
$$

Orthogonal Projection

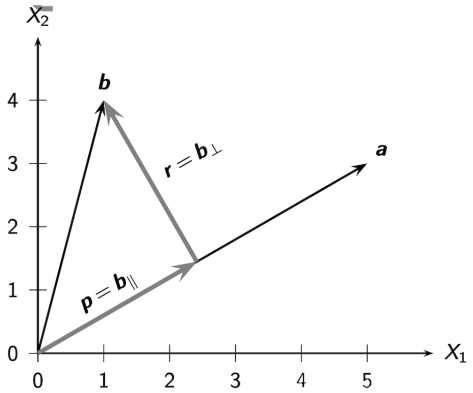


Angle between Vectors

Suppose we have vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ similar to before, but we now wish to calculate the $\cos(\theta)$ where θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.

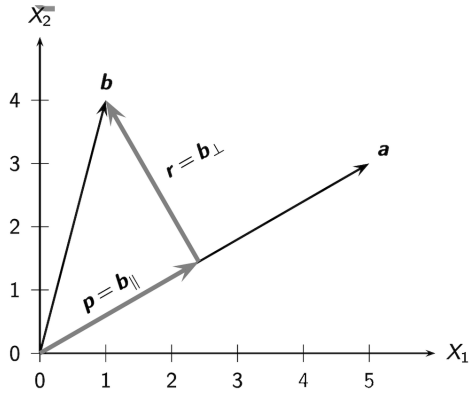


We can think about this problem in terms of the right triangle formed by the projection of $\boldsymbol{b}$ onto $\boldsymbol{a}$:



Angle between Vectors

Now, we treat this problem as finding the $\cos(\theta)$ where θ is the angle between $\boldsymbol{p}$ and $\boldsymbol{b}$



$$\cos(\theta) = \frac{||\boldsymbol{p}||}{||\boldsymbol{b}||}$$

Angle between Vectors

Since we have:

$$\boldsymbol{p} = c\boldsymbol{a} = \left(\frac{\boldsymbol{a}^T\boldsymbol{a}}{\boldsymbol{a}^T\boldsymbol{a}}\right)\boldsymbol{a}$$

and since c is a scalar, we have that:

$$\|\boldsymbol{p}\| = c||\boldsymbol{a}|| = \frac{\boldsymbol{a}^T\boldsymbol{b}}{\boldsymbol{a}^T\boldsymbol{a}}||\boldsymbol{a}|| = \frac{\boldsymbol{a}^T\boldsymbol{b}}{||\boldsymbol{a}||^2}||\boldsymbol{a}|| = \frac{\boldsymbol{a}^T\boldsymbol{b}}{||\boldsymbol{a}||}$$

Thus, we have:

$$\boxed{\cos(\theta) = \frac{||\boldsymbol{p}||}{||\boldsymbol{b}||} = \frac{\boldsymbol{a}^T\boldsymbol{b}}{\|\boldsymbol{a}\|\,\|\boldsymbol{b}\|}}$$

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Random experiment

Definition

A **random experiment** is a process in which some *random* outcome is observed.

Flip a coin

Example: flip a coin. Some random outcome observed - either H or T will be the result, but unknown until after the experiment is complete.

Sample Space

Definition

The **sample space** for an experiment is the set of all possible outcomes for an experiement.

Flip 2 coins

Example: flip 2 coins. Sample space: {(H,H), (H,T), (T,H), (T,T)}

Random Variable

Definition

A *random variable* X is a function whose domain is the sample space. That is, random variable, $X : \mathcal{O} \to \mathbb{R}$, where $\mathcal{O}$ is the set of all possible outcomes of the experiment.

Flip 2 coins

Example: Suppose X is a random variable representing the number of H on a flip of 2 coins. Then we have the following:

Outcome	X
(H,H)	2
(H,T)	1
(T,H)	1
(T,T)	0

Discrete Random Variable

Definition

A *discrete random variable* takes on only a finite or countably infinite number of values, whereas a *continuous random variable* if it can take on any value in $\mathbb{R}$.

Discrete Variable: Long Sepal Length

Define random variable A, denoting long sepal length (7cm or more) as follows:

$$A(v) = \begin{cases} 0 & \text{if } v < 7 \\ 1 & \text{if } v \geq 7 \end{cases}$$

The sample space of A is $\mathcal{O} = [4.3, 7.9]$, and its range is $\{0, 1\}$. Thus, A is discrete.

Discrete Random Variable

By default, a numeric attribute X_j is considered as the identity random variable given as

$$X(v) = v$$

$\forall v \in \mathcal{O}$. Here $\mathcal{O} = \mathbb{R}$.

Person's Height

Define X as a random variable representing a person's height, h.

$$X(h) = h$$

$\forall h \in \mathcal{O}$.

Probability Mass Function

Definition

Suppose X is a discrete random variable. Then, the **probability mass function**, f, of X is defined as

$$f(x) = P(X = x) \qquad \forall x \in \mathbb{R}$$

f must obey the basic rules of probability. That is, f must be non-negative:

$$f(x) \geq 0$$

and the sum of all probabilities should add to 1:

$$\sum_x f(x) = 1$$

Intuitively, for a discrete variable X, the probability is concentrated or massed at only discrete values in the range of X, and is zero for all other values.

Bernoulli Distribution

Definition

Suppose we have a random experiment consisting of a single trial having two possible outcomes: success or failure, where $p(\text{success}) = p$ and $p(\text{failure}) = 1 - p$.

Suppose we have a random variable, X whose values are defined as follows:

$$X = \begin{cases} 1 & \text{if success} \\ 0 & \text{if failure} \end{cases}$$

Then, X has a **Bernoulli Distribution** given by the following:

$$p(X = 1) = p$$
$$p(X = 0) = 1 - p$$

Example: Iris Dataset Extract - Sepal Length

Iris Dataset Extract: `sepal length` (in centimeters)

5.9	6.9	6.6	4.6	6.0	4.7	6.5	5.8	6.7	6.7	5.1	5.1	5.7	6.1	4.9
5.0	5.0	5.7	5.0	7.2	5.9	6.5	5.7	5.5	4.9	5.0	5.5	4.6	7.2	6.8
5.4	5.0	5.7	5.8	5.1	5.6	5.8	5.1	6.3	6.3	5.6	6.1	6.8	7.3	5.6
4.8	7.1	5.7	5.3	5.7	5.7	5.6	4.4	6.3	5.4	6.3	6.9	7.7	6.1	5.6
6.1	6.4	5.0	5.1	5.6	5.4	5.8	4.9	4.6	5.2	7.9	7.7	6.1	5.5	4.6
4.7	4.4	6.2	4.8	6.0	6.2	5.0	6.4	6.3	6.7	5.0	5.9	6.7	5.4	6.3
4.8	4.4	6.4	6.2	6.0	7.4	4.9	7.0	5.5	6.3	6.8	6.1	6.5	6.7	6.7
4.8	4.9	6.9	4.5	4.3	5.2	5.0	6.4	5.2	5.8	5.5	7.6	6.3	6.4	6.3
5.8	5.0	6.7	6.0	5.1	4.8	5.7	5.1	6.6	6.4	5.2	6.4	7.7	5.8	4.9
5.4	5.1	6.0	6.5	5.5	7.2	6.9	6.2	6.5	6.0	5.4	5.5	6.7	7.7	5.1

Define random variable A as follows: $A(v) = \begin{cases} 0 & \text{if } v < 7 \\ 1 & \text{if } v \geq 7 \end{cases}$

We find that only 13 Irises have sepal length of at least 7 cm. Thus, the probability mass function of A can be estimated as:

$$f(1) = P(A = 1) = \frac{13}{150} = 0.087 = p$$

and

$$f(0) = P(A = 0) = \frac{137}{150} = 0.913 = 1 - p$$

A has a *Bernoulli distribution* with parameter $p \in [0, 1]$, which is the probability of picking an Iris with a long sepal length.

Binomial Distribution

Definition

Suppose we have a random experiment consisting of n independent trials, each of which has two possible outcomes - success and failure - where the $p(\text{success}) = p$ and $p(\text{failure}) = 1 - p$.

Suppose X is a random variable denoting the number of successes, k, among the trials in this experiment. Then, X has a **Binomial Distribution** whose probability mass function, (also known as probability distribution), is given by:

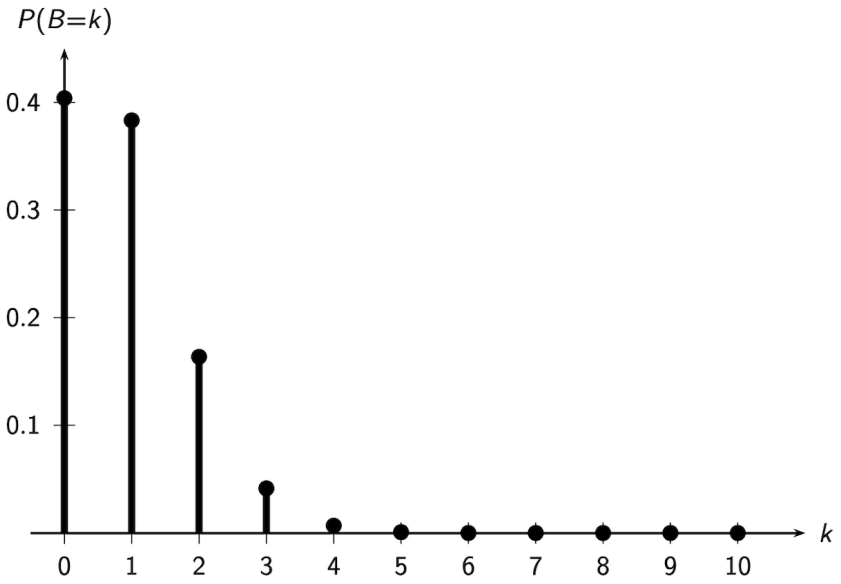
$$f(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Example: Iris Dataset Extract - Sepal Length

Define discrete random variable B, denoting the number of Irises with long sepal length in n independent Bernoulli trials with probability of success $p = 0.087$.

B is a Binomial Random Variable on the discrete interval $[0, n]$:

$$f(k) = P(B = k) = \binom{n}{k} p^k (1-p)^{m-k}$$



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Probability Density Function

Definition

Suppose X is continuous random variable. The **probability density function** of X is defined as

$$P\big(X \in [a, b]\big) = \int_a^b f(x)\ dx$$

f must obey the basic rules of probability. That is, f must be non-negative:

$$f(x) \geq 0$$

and the sum of all probabilities should add to 1:

$$\int_{-\infty}^{\infty} f(x)\ dx = 1$$

Probability Density Function

- Note that $P(X = v) = 0 \ \forall v \in \mathbb{R}$ since there are infinite possible values in the sample space.
- This means is that the probability mass is spread so thinly over the range of values that it can be measured only over intervals $[a, b] \subset \mathbb{R}$, rather than at specific points.

Normal Distribution

Definition

Suppose X is continuous random variable. X has a **Gaussian** or **normal distribution** if its probability density function is given by the following:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\frac{-(x-\mu)^2}{2\sigma^2}\right\}$$

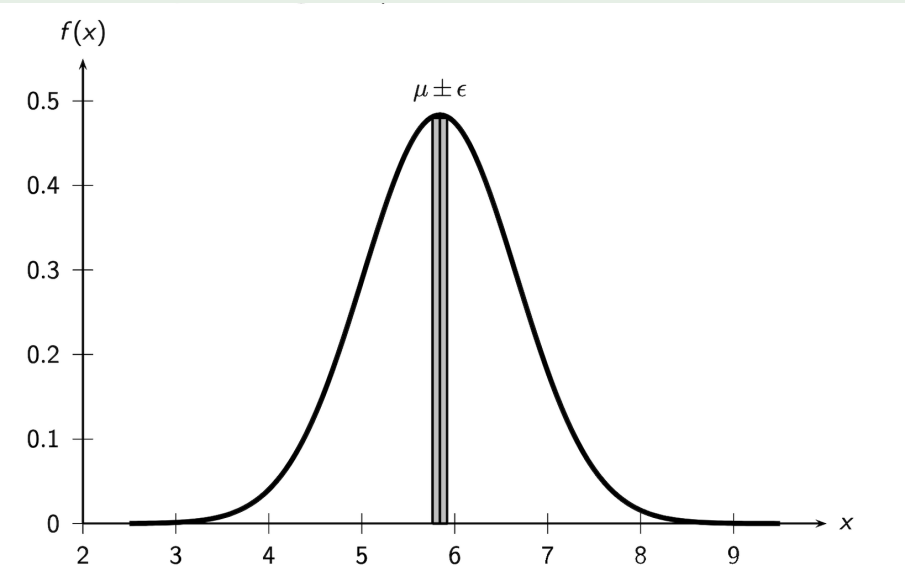
where $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$ is the mean value, and $\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$ is the variance.

Example: Iris Dataset Extract: Sepal Length

Example: Iris Dataset Extract: Sepal Length

We model sepal length via the *Gaussian* density function as follows:

Normal distribution for sepal length: $\mu = 5.84$, $\sigma^2 = 0.681$



Cumulative Distribution Function

Definition

For random variable X, its **cumulative distribution function (CDF)** $F : \mathbb{R} \rightarrow [0, 1]$, gives the probability of observing a value less than or equal to a given value, x:

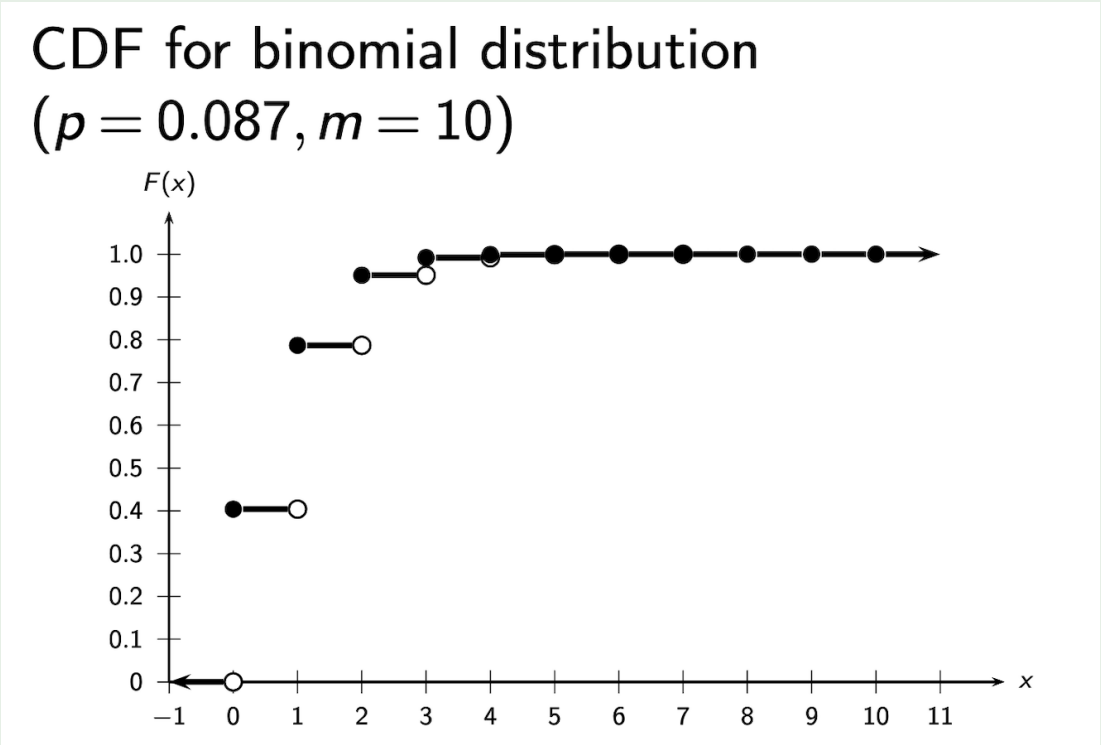
$$F(x) = P(X \leq x) \qquad \forall x | -\infty < x < \infty$$

Cumulative Distribution Function: Discrete Case

When X is discrete, F is given as

$$F(x) = P(X \leq x) = \sum_{u \leq x} f(u)$$

Iris Dataset Extract: Sepal Length

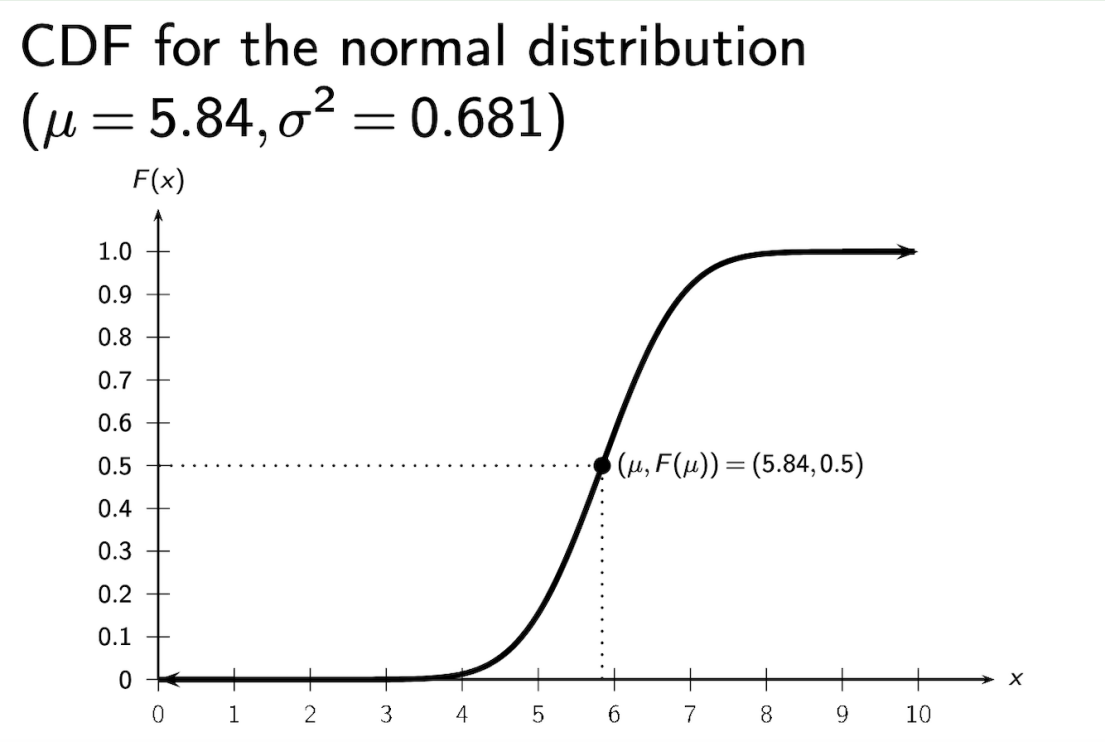


Cumulative Distribution Function: Continuous Case

When X is continuous, F is given as

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u)\ du$$

Iris Dataset Extract: Sepal Length



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Bivariate Distribution: Discrete Case

Iris Dataset Extract: Sepal Length, Sepal Width

Define discrete random variables

$$\text{long sepal length: } X_1(v) = \begin{cases} 1 & \text{if } v \geq 7 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{long sepal width: } X_2(v) = \begin{cases} 1 & \text{if } v \geq 3.5 \\ 0 & \text{otherwise} \end{cases}$$

The bivariate random variable

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

has the joint probability mass function

$$f(\boldsymbol{x}) = P(\boldsymbol{X} = \boldsymbol{x})$$

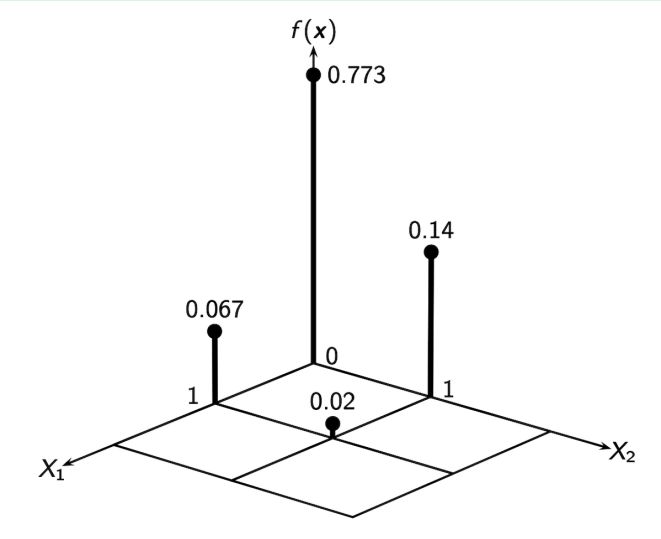
$$\text{i.e., } f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$$

Bivariate Distribution: Discrete Case

Iris Dataset Extract: Sepal Length, Sepal Width

Iris: joint PMF for long sepal length and sepal width

$$f(0,0) = P(X_1 = 0, X_2 = 0) = 116/150 = 0.773$$
$$f(0,1) = P(X_1 = 0, X_2 = 1) = 21/150 = 0.140$$
$$f(1,0) = P(X_1 = 1, X_2 = 0) = 10/150 = 0.067$$
$$f(1,1) = P(X_1 = 1, X_2 = 1) = 3/150 = 0.020$$



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Bivariate Distribution: Continuous Case

Iris Dataset Extract: Sepal Length and Sepal Width

Bivariate Normal: modeling joint distribution for sepal length (X_1) and sepal width (X_2)

$$f(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{2\pi\sqrt{|\boldsymbol{\Sigma}|}} \quad \exp\left\{-\frac{(\boldsymbol{x}-\boldsymbol{\mu})^T \, \boldsymbol{\Sigma}^{-1} \, (\boldsymbol{x}-\boldsymbol{\mu})}{2}\right\}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ specify the 2D mean and covariance matrices:

$$\boldsymbol{\mu} = (\mu_1, \mu_2)^T \qquad\qquad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}$$

with mean $\mu_i = \frac{1}{n}\sum_{k=1}^{n} x_{ki}$ and covariance $\sigma_{ij} = \frac{1}{n}\sum_{k=1}(x_{ki}-\mu_i)(x_{kj}-\mu_j)$. Also, $\sigma_i^2 = \sigma_{ii}$.

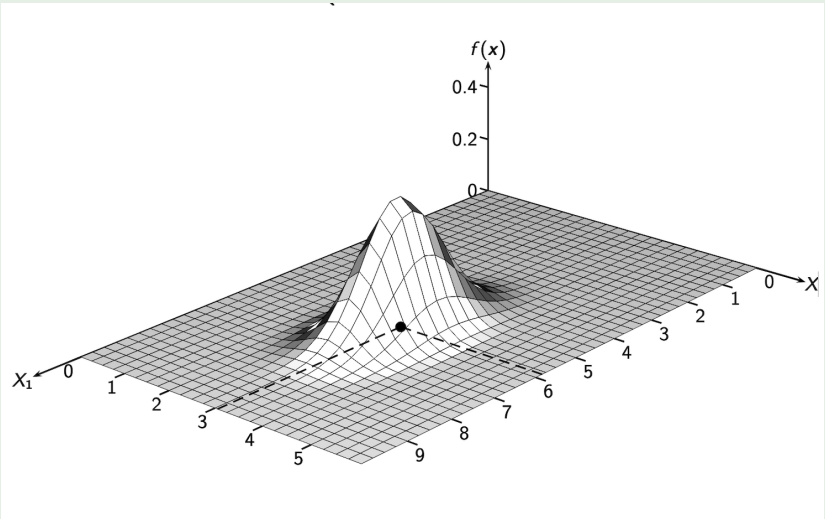
Bivariate Distribution: Continuous Case

Iris Dataset Extract: Sepal Length and Sepal Width

Bivariate Normal

$$\boldsymbol{\mu} = (5.843, 3.054)^T$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} 0.681 & -0.039 \\ -0.039 & 0.187 \end{pmatrix}$$



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Independence

Two random variables X_1 and X_2 are (statistically) *independent* if, for every $W_1 \subset \mathbb{R}$ and $W_2 \subset \mathbb{R}$, we have

$$P(X_1 \in W_1 \text{ and } X_2 \in W_2) = P(X_1 \in W_1) \cdot P(X_2 \in W_2)$$

which also implies that

$$F(\boldsymbol{x}) = F(x_1, x_2) = F_1(x_1) \cdot F_2(x_2)$$
$$f(\boldsymbol{x}) = f(x_1, x_2) = f_1(x_1) \cdot f_2(x_2)$$

where F_i is the cumulative distribution function, and f_i is the probability mass or density function for random variable X_i.

Random Sample

Given a random variable X, a *random sample* of size n from X is defined as a set of n *independent and identically distributed (IID)* random variables

$$S_1, S_2, \ldots, S_n$$

The S_i's have the same probability distribution as X, and are statistically independent.

Multivariate Random Sample

Given dataset $\boldsymbol{D}$, the n data points $\boldsymbol{x}_i$ (with $1 \leq i \leq n$) constitute a d-dimensional *multivariate random sample*. That is, we have n independent observations of the d-dimensional random variable, $\boldsymbol{X} = (X_1, X_2, \ldots, X_d)$.

Since the $\boldsymbol{x}_i$ are assumed to be independent and identically distributed, their joint distribution is given as

$$f(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n) = \prod_{i=1}^{n} f_{\boldsymbol{X}}(\boldsymbol{x}_i)$$

where $f_{\boldsymbol{X}}$ is the probability mass or density function for $\boldsymbol{X}$.

Assuming that the d attributes $X_1, X_2, \ldots, X_d$ are statistically independent, the joint distribution for the entire dataset is given as:

$$f(\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n) = \prod_{i=1}^{n} f(\boldsymbol{x}_i) = \prod_{i=1}^{n} \prod_{j=1}^{d} f_{X_j}(x_{ij})$$

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Inference Statistic

Let $\{\boldsymbol{S}_i\}_{i=1}^m$ be a random sample of size m drawn from a (multivariate) random variable $\boldsymbol{X}$. A *statistic* $\hat{\theta}$ is a function

$$\hat{\theta} \colon (\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_m) \to \mathbb{R}$$

The statistic is an estimate of the corresponding population parameter θ, where the *population* refers to the entire universe of entities under study. *The statistic is itself a random variable.*

Inference Statistic: Sample Mean

Definition

The **sample mean** is a statistic defined as the average of the features across a sample:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Example: Iris Dataset Extract: Sample Mean

For `sepal length`, we have $\hat{\mu} = 5.84$, which is an estimator for the (unknown) true population mean sepal length.

Inference Statistic: Sample Variance

Definition

The **sample variance** is an inference statistic given by the following:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

Example: Iris Dataset Extract: Sample Variance

For sepal length, we have $\hat{\sigma}^2 = 0.681$.

Inference Statistic: Total Variance

Definition

The **total variance** is a multivariate statistic given by:

$$var(\boldsymbol{D}) = \frac{1}{n}\sum_{i=1}^{n}\|\boldsymbol{x}_i - \boldsymbol{\mu}\|^2$$

Example: Iris Dataset Extract: Total Variance

For the Iris data (with 4 attributes: sepal length and width, petal length and width), we have $var(\boldsymbol{D}) = 0.868$.